

Beixuan Dong

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Webpage: <https://scholar.google.com/citations?user=VRLxkWYAAAAJ&hl=en>

Education Background

University of Alberta (UofA), Edmonton, Canada	09/2023-present
PhD student in Department of Mechanical Engineering, GPA: 3.8/4.0	
Supervisor: Dr. Xinming Li Co-supervisor: Dr. Lingzi Wu	
Central South University (CSU), Changsha, China	09/2020-06/2023
MSc in Department of Engineering Management, GPA: 3.6/4.0	
Supervisor: Dr. Ming Shan	
Central South University of Forestry and Technology (CSUFT), Changsha, China	09/2016-06/2020
B. Eng in Engineering Management GPA: 4.0/4.0 Ranking: 1/73	

Academic Experience

University of Washington (UW), Seattle, USA	06/2025-08/2025
Visiting PhD in Department of Construction Management	
Supervisor: Dr. Lingzi Wu	
National University of Singapore (NUS), Singapore	12/2021-07/2022
Visiting Master in Department of Built Environment	
Supervisor: Dr. Bon-Gang Hwang	

Research Interests

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- Infrastructure resilience
 - Data-driven modeling
 - AI application in disaster management

Research Experience

Data-Driven Modeling for Disaster Resilience in Road Networks: Impact Assessment, Prediction, and Restoration
2023-present

Advised by Dr. Xinming Li and Dr. Lingzi Wu

- Systematically reviewed studies on natural disaster impacts, focusing on the capability and applicability of the methods used in these studies.

- Developed a GNN-Based framework that incorporated network structural, functional, and social features to assess and predict the flood impacts on highway networks.
- Proposed a GNN-based framework that prioritizes road restoration by considering both the functional importance of roads and the access needs of socially vulnerable communities.
- To formulate post-disaster road restoration as an edge-level, resource-constrained scheduling problem, where each decision aims to maximize the marginal accessibility gain while accounting for dynamic social vulnerability.

Assessment, Simulation, and Improvement of Resilience in Urban Rail Transit Construction Projects 2020-2023

Advised by Dr. Ming Shan and Dr. Bon-Gang Hwang

- Identifying key attributes of physical resilience of urban rail transit construction projects and the mental resilience of contractors in urban rail transit construction projects.
- Developing a multi-agent-based model in AnyLogic that can simulate the resilience of urban rail transit construction projects facing disruptive events, particularly the complex interactions among disruptive events, physical performance of urban rail transit infrastructures, and mental well-being of project stakeholders.
- Applying the proposed model in a real-world case for verification and proposing strategies for resilience improvement and etc.

Publications

Journal papers

- [1]. **Dong, B.**, Wu, L. & Li, X. (2026). Towards equitable post-disaster highway restoration: A GNN-based framework for segment-level prioritization. *Transportation Research Part D: Transport and Environment*. (under review, IF: 8.7)
- [2]. **Dong, B.**, Li, X., & Wu, L. (2026). A GNN-Based Framework for Assessing Flood Impacts on Highway Networks: Integrating Network Structural, Functional, and Social Features. *Journal of Management in Engineering*, 42(4), 04026010. (IF: 7)
- [3]. **Dong, B.**, Ding, S., Wu, L., & Li, X. (2025). Short-term natural disaster impacts on transportation infrastructure: a systematic review. *Natural Hazards*, 121(15), 17321-17362. (IF: 4.5)
- [4]. **Dong, B. X.**, Shan, M., Cui, J. H., & Hwang, B. G. (2025). Examining mental resilience of contractor in urban rail transit construction project under disruptive events. *Tunnelling and Underground Space Technology*, 163, 106725. (IF: 8.7)
- [5]. **Dong, B. X.**, Shan, M., & Hwang, B. G. (2022). Simulation of transportation infrastructures resilience: a comprehensive review. *Environmental Science and Pollution Research*, 29(9), 12965-12983.

Conference papers

- [1]. **Dong, B.**, Wu, L., & Li, X. Flood Impact Assessment on Highway Networks Considering Socioeconomic Vulnerability: Evidence from Harris County. 43rd International Symposium on Automation and Robotics in Construction (ISARC), June 22-26, 2026, Singapore. (Accepted)
- [2]. **Dong, B.**, Zhou, C., Gue, B., Li, X., & Wu, L. Impacts Identification of Supplier Reliability on Project Duration in Heavy Industrial Construction Supply Chain Using Discrete Event Simulation. The 10th Joint CRC/CSCE Construction Speciality Conference, 28-31st July, 2025, Montreal, Quebec, Canada
- [3]. **Dong, B.**, Wu, L., & Li, X. Short-Term Disaster Impacts on Transportation Infrastructure: A Review of

Emerging Technologies and Approaches. Computing in Civil Engineering, July 28–31, 2024, Pittsburgh, Pennsylvania, USA

Conference and poster presentation

- [1]. **Dong, B. X.** Flood Impact Assessment on Highway Networks Considering Socioeconomic Vulnerability: Evidence from Harris County. 43rd International Symposium on Automation and Robotics in Construction (ISARC), June 22-26, 2026, Singapore
- [2]. **Dong, B.,** Zhou, C., Gue, B., Li, X., & Wu, L. Impacts Identification of Supplier Reliability on Project Duration in Heavy Industrial Construction Supply Chain Using Discrete Event Simulation. The 10th Joint CRC/CSCE Construction Speciality Conference, 28-31st July, 2025, Montreal, Quebec, Canada
- [3]. **Dong, B.** Impacts Identification of Supplier Reliability on Project Duration in Heavy Industrial Construction Supply Chain Using Discrete Event Simulation [Poster presentation], Construction Innovation Center (CIC), June 12-13, 2024, Edmonton, Canada.
- [4]. **Dong, B.** Short-Term Disaster Impacts on Transportation Infrastructure: A Review of Emerging Technologies and Approaches. Computing in Civil Engineering, July 28–31, 2024, Pittsburgh, Pennsylvania, USA
- [5]. **Dong, B.** A Comprehensive Review of Simulation Methods Used in Research of Transportation Infrastructures Resilience”, 27th International Symposium on Advancement of Construction Management and Real Estate, 5-6 Dec. 2022, Hong Kong, China

Dissertation

- [1]. **Dong, B.X.** (2023) Multi-agent-based Simulation of Resilience in Urban Rail Transit Construction Projects, M.S. Dissertation, Central South University, Changsha, China

Book & Book Chapter

- [1]. Yu D.M., Shan M., Li Y.G. (2021). International Railway Project Risk Management. Central South University Press.

Note: As a helper to draft Chapter 12 of the book, which is Construction Insurance and Surety Bonds.

Awards & Scholarships

- [1]. Academic Travel Grant, UofA, 500 CAD, 2026
- [2]. Alberta Innovated Graduate Scholarship, UofA, 24000 CAD, 2025-2027
- [3]. Academic Travel Grant, UofA, 500 CAD, 2025
- [4]. Second Class Merit Based Scholarship, CSU, 2021&2022
- [5]. First Class Merit Based Scholarship, CSU, 2020
- [6]. Outstanding Undergraduate Graduates of Hunan Province, China, 2020 (Top 5%)
- [7]. National Scholarship for Undergraduate Students, China, 8000 CNY, 2019 (Top 1%)
- [8]. Third Class Mathematical Contest in Modeling of Hunan Province, China, 2018
- [9]. First Class Merit Based Scholarship, CSUFT, 3000 CNY, 2016-2018 (Top 5%)

[10]. School-level “Teacher’s Grace and Zhu Yuxuan” Scholarship, CSUFT, 1000 CNY, 2017 (Top 5%)

[11]. School-level Merit Student, CSUFT, 2016 (Top 5%)

Teaching Experience

ENG M 632: Project Risk Management, University of Alberta 01/2026-04/2026

- Answering student questions
- Grading assignment

ENG M 620: Engineering Economic Analysis, University of Alberta 01/2026-04/2026

- Answering student questions
- Grading assignment

ENG M 530: Engineering Project Management, University of Alberta 09/2025-12/2025

- Assist in editing assignment questions
- Answering student questions
- Grading assignment

ENG M 530: Engineering Project Management, University of Alberta 05/2024-07/2024

- Independently teaching one course: project schedule management
- Answering student questions
- Grading assignment

Professional Services

Professional affiliations

Student Member – American Society of Civil Engineers (ASCE)

Reviewer Services

2024 - present	ASCE-Journal of Construction Engineering and Management
2026	Elsevier-Cities
2026	Elsevier-Clean Cities
2026	43rd International Symposium on Automation and Robotics in Construction, Singapore
2026	CI & CRC Joint Conference, San Antonio, USA
2025	42rd International Symposium on Automation and Robotics in Construction, Montreal, Canada
2024	41rd International Symposium on Automation and Robotics in Construction, Lille, France

Skills & Languages

Languages: Chinese (Native), English (Professional working proficiency)

Research Skills: English Publish Presentation, Multi-tasking, Team Work, Information Retrieval, English Essay Writing and Revision Skills

Technical Skills: Python, AnyLogic, Autodesk Revit, Java, C++, Matlab